

Toward Testing the One-Vote-Veto Hypothesis in Overall IEQ Satisfaction in Belgian Classrooms

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Abstract

This paper investigates whether overall indoor environmental quality (IEQ) satisfaction in classrooms can be predicted from environmental, contextual, occupant, and building-related variables. The analysis uses an open Belgian classroom dataset containing 6,834 repeated satisfaction assessments from 321 occupants across primary, secondary, and university classrooms. The original five-point overall IEQ scale is collapsed into three classes: dissatisfied, neutral, and satisfied. Five model families are compared under matched five-fold cross-validation, using macro F1 as the main selection metric because the target distribution is imbalanced. The best single model is an Extra Trees classifier, reaching accuracy = 0.722, balanced accuracy = 0.592, macro F1 = 0.591, weighted F1 = 0.719, and ordinal MAE = 0.316.

The confusion matrices show that satisfied votes are predicted most reliably, while dissatisfied and neutral responses remain harder to separate. Feature-importance and sensitivity analyses suggest that overall IEQ satisfaction is shaped by a combination of environmental, occupant, and contextual factors rather than by one dominant domain alone. Follow-up experiments with complete-case filtering, feature engineering, synthetic class balancing, and primary-school-specific training produce only modest gains and reveal domain shift between educational settings. The results quantify the performance and limits of classroom IEQ prediction and show how such models could support occupant-centered and energy-aware decision support in school buildings.

Keywords: indoor environmental quality; classroom comfort; occupant satisfaction; machine learning; Extra Trees; transfer learning; Belgian classroom dataset

1 Introduction

Indoor environmental quality (IEQ) describes the combined indoor conditions experienced by occupants, especially thermal comfort, indoor air quality, lighting, and acoustics. In classrooms, these conditions can affect students' comfort, engagement, well-being, and learning [1, 2, 3, 4, 5]. These effects are large enough to matter in everyday classroom operation, and they are not limited to obviously uncomfortable rooms. In a field study of 140 fifth-grade classrooms, each 1 L/s per person increase in ventilation within 0.9–7.1 L/s per person was associated with an increase of up to about 0.5% in mathematics scores. Within the ordinary 20–25 °C classroom-temperature range, each 1 °C decrease was associated with a further increase of about 0.5–0.6%. Published school-specific relationships also suggest larger effects under poorer conditions: lowering classroom temperature from 30 °C to 20 °C can increase performance by about 20%, and reducing classroom CO₂ from 2,100 ppm to 900 ppm can improve task speed by about 12% [6, 7, 8]. Recent thermal-comfort research reflects this need: PMV and adaptive comfort models are increasingly complemented by personal comfort models that use IoT data and machine learning to predict individual thermal preference [9, 10, 11]. These models are also moving into practice: personal-comfort-based control has been tested in occupied buildings, and one commercial system has used machine learning to translate warm/cool feedback into HVAC commands and thermostat schedules [12, 13]. However, this progress still focuses mainly on thermal comfort and HVAC control, not on overall classroom IEQ across air quality, lighting, and acoustics. In practice, classrooms are often monitored through separate indicators such as temperature, CO₂, illuminance, and sound level. These separate limits do not show whether students perceive the room as acceptable overall. They also miss operational trade-offs: shading can reduce overheating but lower daylight, and more ventilation can reduce CO₂ but affect thermal comfort, noise, and energy use [14, 15, 16, 17]. This paper therefore focuses on developing a model for predicting overall IEQ satisfaction: how students judge the

classroom environment as a whole, rather than as isolated responses to individual IEQ domains. The aim is to support better handling of energy-comfort trade-offs in classroom operation [18, 19, 20].

Several gaps in the IEQ literature motivate this focus. Reviews show that overall IEQ satisfaction is affected by several domains at the same time, but many existing models do not fully capture these interactions and have only limited validation across real buildings [15, 16, 20, 17]. Many models also rely on subjective votes for separate domains, such as thermal, air-quality, visual, or acoustic satisfaction. These inputs are useful for research, but they are not always available in routine classroom monitoring, where decisions must often rely on sensor and context data. Recent sensor-based studies show that monitored environmental data can predict perceived IEQ, including student perceptions in schools [21, 22, 5]. However, classrooms remain less developed than offices or commercial buildings: students have limited control over the indoor environment, may respond differently from adults, and existing studies are difficult to compare or reuse [23, 24, 25, 26]. The open Belgian classroom dataset helps address this gap by linking repeated overall and domain satisfaction votes with environmental, personal, and contextual variables across primary, secondary, and university classrooms [27]. This paper uses that dataset to test how well overall IEQ satisfaction can be predicted from classroom sensor, occupant, and context data, whether the prediction can be personalized, and how well models generalize across educational settings.

1.1 Research Questions

1. **RQ1: What prediction accuracy can be achieved for IEQ satisfaction using sensor and contextual data?**
2. **RQ2: Which sensor, occupant, and contextual variables are most influential to IEQ satisfaction?**
3. **RQ3: How many occupant votes are required to achieve effective personalization for individual occupants or specific classrooms?**
4. **RQ4: What performance loss occurs when prediction models are transferred across educational settings?**

1.2 Contribution

The practical value of this work is to move classroom IEQ assessment from isolated threshold checking toward occupant-centered and energy-aware decision support. Instead of evaluating temperature, CO₂, light, or sound separately, the study asks whether available classroom data can indicate when students are likely to perceive the indoor environment negatively as a whole.

- It provides a first step toward future energy-comfort optimization by predicting overall IEQ satisfaction instead of treating temperature, CO₂, light, and sound as separate thresholds.
- It evaluates how well a three-class model can predict overall IEQ satisfaction and identifies where the prediction remains uncertain (RQ1).
- It identifies which measured variables are most useful for dashboard warnings and intervention prioritization (RQ2).
- It tests whether prediction can be improved by adapting the pooled model to a specific classroom or occupant (RQ3).
- It evaluates cross-setting generalization by testing whether models trained in one educational setting remain reliable in another; if an independent classroom dataset becomes available, this step will be extended to external transfer-learning validation (RQ4).

2 Dataset Description

The source material is the open Belgian classroom dataset with 6,834 satisfaction assessments from 321 occupants across 11 classrooms in three educational settings: primary school, secondary school, and a university test lecture

room [27]. It was collected in Flanders, Belgium: in three secondary-school classrooms in Haacht from April to June 2021, seven primary-school classrooms in five schools from October to December 2021, and a test lecture room at KU Leuven Ghent Campus from November 2021 to May 2022 [27]. The secondary-school campaign covered intermediate-season conditions, the primary-school campaign covered heating-season conditions, and the test lecture room covered heating and intermediate-season conditions [27].

Occupant feedback was collected through repeated right-here-right-now online surveys. The surveys recorded five-point satisfaction votes for thermal, indoor-air-quality, acoustic, visual, and overall IEQ conditions, together with personal and contextual information such as occupant ID, age, gender, clothing, seating position, survey moment, activity, and classroom or group; each vote was linked to the nearest environmental measurements in time [27].

The environmental monitoring setup differed between the three campaigns. In the secondary school, three Renson Sense IEQ devices were placed in each classroom and measured air temperature, relative humidity, CO₂, light intensity, and sound level at 5-minute intervals; outdoor temperature was measured separately [27]. In the primary schools, classroom IEQ was monitored using HOBO-MX, HOBO U12-O12, and Netatmo smart weather station devices, also at 5-minute intervals, with an additional CO₂ monitoring device in each classroom to inform pupils and teachers about IAQ conditions [27]. In the test lecture room, building sensors and a weather station measured indoor and outdoor environmental conditions at 1-minute intervals, an Ethern NEMO IAQ sensor measured IAQ variables at 10-minute intervals, and a Testo 480 thermal comfort meter measured additional thermal variables such as air velocity and mean radiant temperature [27].

This study uses the combined simplified version of the dataset, which keeps variables available across the three cases. It includes the original five-point overall IEQ satisfaction vote and a common set of environmental, occupant, and contextual variables, such as air temperature, relative humidity, CO₂, illuminance, LAeq, clothing, age, gender, seating position, classroom type, and survey timing.

Supervisor note: The following graphs are here only to help understanding the dataset.

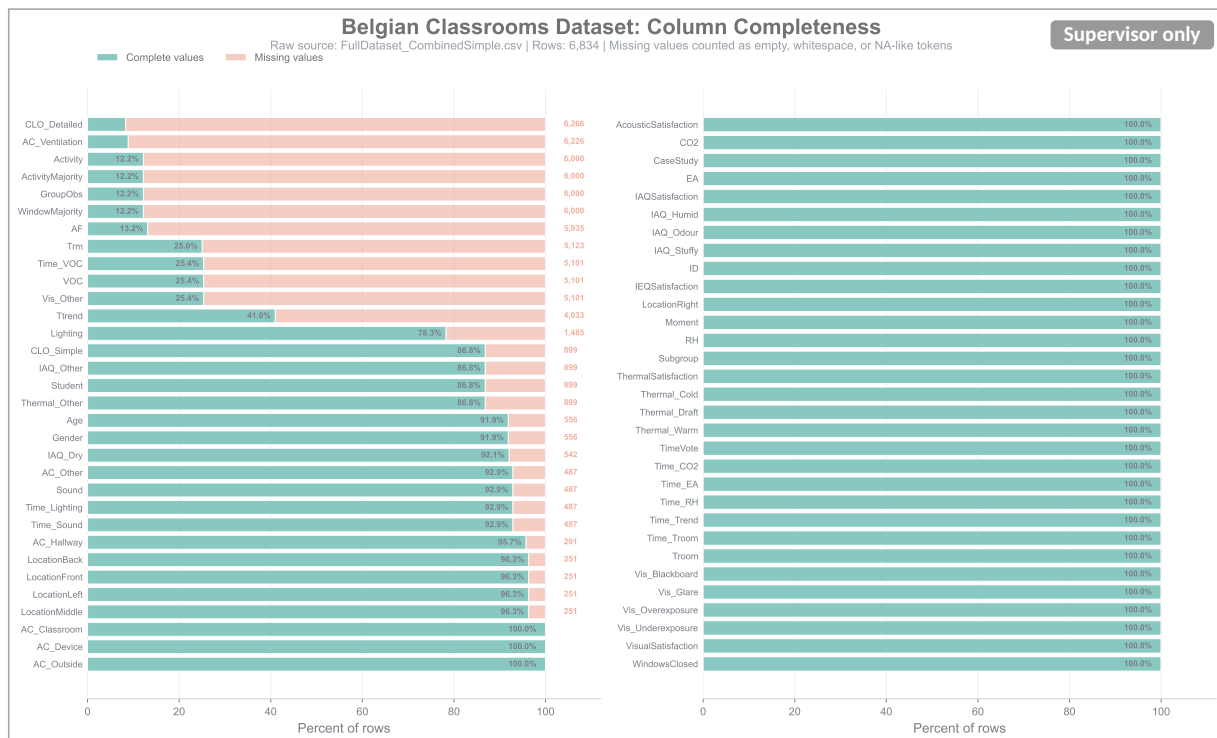


Figure 1: Supervisor-only supporting figure: raw dataset column completeness across the shared classroom dataset.

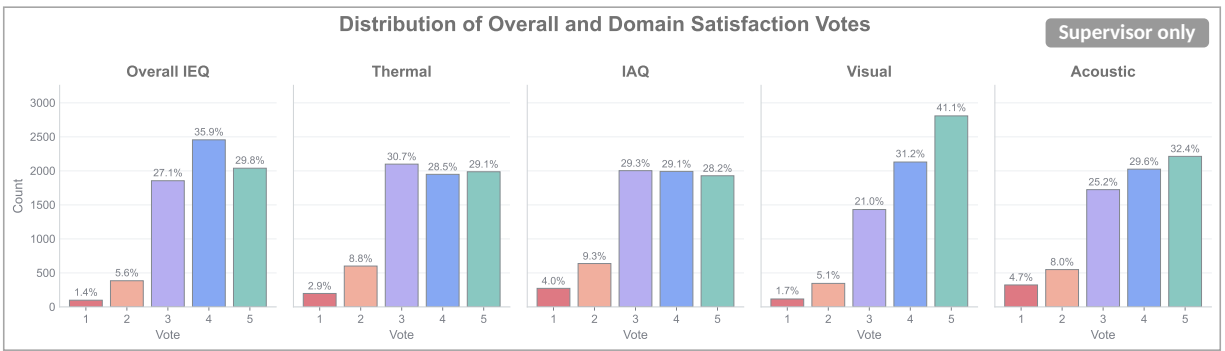


Figure 2: Supervisor-only supporting figure: satisfaction vote distributions across the overall IEQ target and the four domain targets on the original five-class scale.

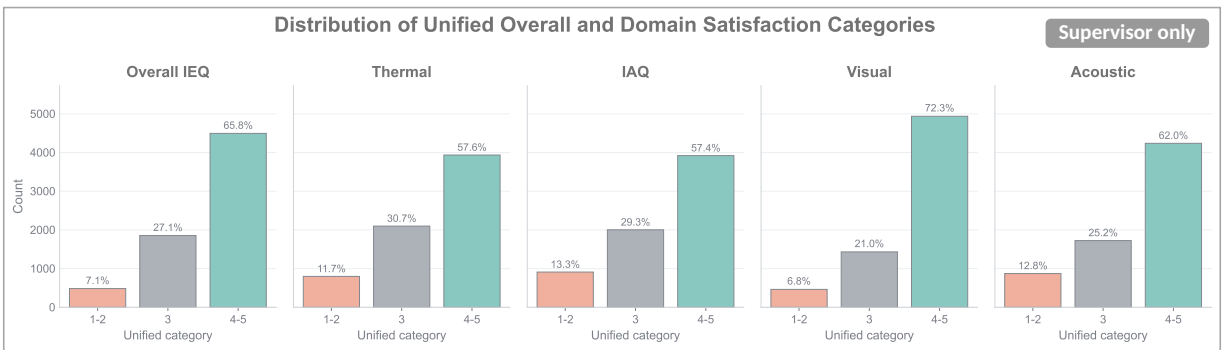


Figure 3: Supervisor-only supporting figure: unified satisfaction-category distributions across the shared target set.

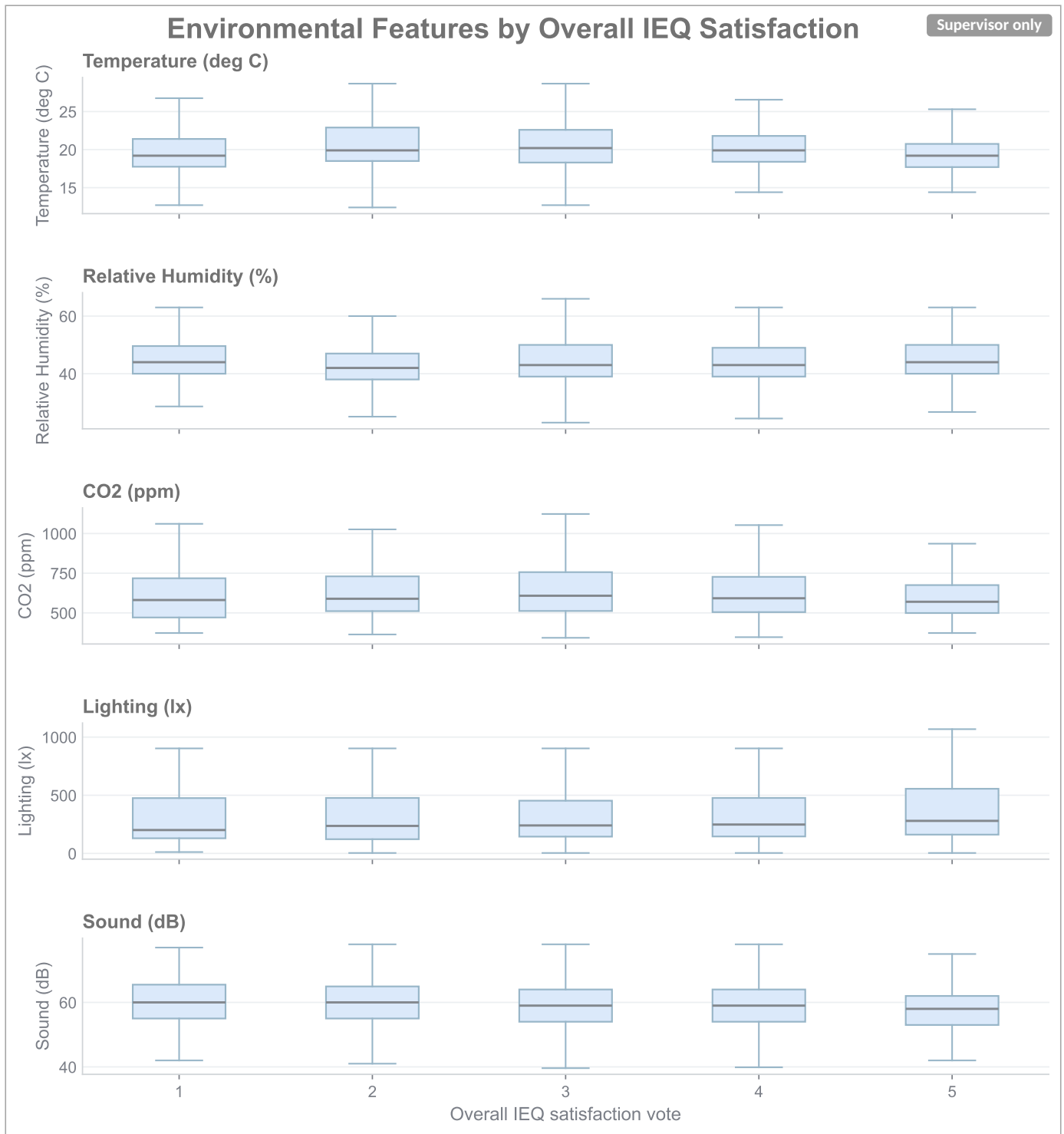


Figure 4: Supervisor-only supporting figure: environmental feature boxplots for the IEQ paper variables.

2.1 Data Cleaning and Imputation

Before imputation, the model input and output variables were defined explicitly. The output variable was the overall IEQ satisfaction vote. The input variables were grouped into sensor-based measurements (Temperature, RH, CO2, Lighting, Sound, and Ttrend), occupant descriptors (Age, Gender, Student, and CL0), seating context (classroom location), and other contextual variables (EA, Vote hour, Vote weekday, Moment, and AC on/off). In the model-ready dataset, classroom location was represented by five binary columns: LocationBack, LocationFront, LocationLeft,

LocationRight, and LocationMiddle. Almost all variables that were consistently available in the combined simplified dataset were retained. The remaining variables were excluded mainly because they were not available across the three cases, had too many missing values for a comparable model, or would introduce leakage. In particular, domain-specific satisfaction votes were not used as predictors because they are subjective responses close to the target variable, while trace and grouping identifiers, including Occupant ID, Group, and Subgroup, were excluded to avoid identity- or classroom-specific leakage and unintended personalization to individual occupants or classrooms. The completeness of the retained inputs is summarized in Figure 5.

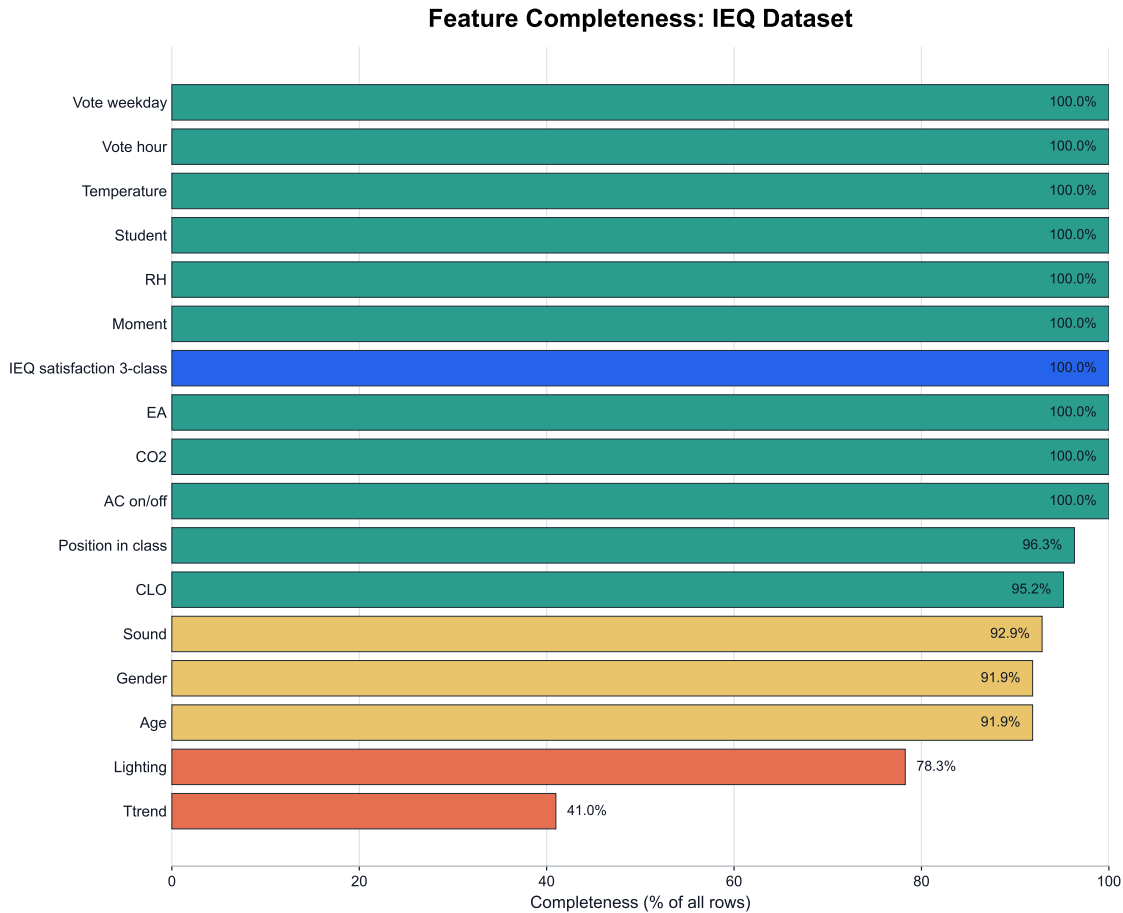


Figure 5: Feature completeness for the variables currently retained in the IEQ paper.

Missing values were filled using fixed, reproducible rules. Missing classroom position was set to Middle. Clothing insulation (CLO) was imputed by the median within each Group, with the global dataset median used as fallback. In the survey and dataset documentation, clothing is recorded as either a simple three-level category (light, neutral, warm) or a more detailed clothing description, not as a directly measured insulation value. Here, the simple categories were mapped to approximate clo values of 0.44, 0.64, and 0.81 based on standard clothing-insulation tables; these values should therefore be interpreted as representative approximations rather than observed measurements [27, 28, 29]. Temperature trend, sound level, and lighting were imputed by the median within Group + Subgroup, then by Group, and finally by the global median as a fallbacks. Age was first inferred from other records with the same Occupant ID using that occupant's median reported age; if no age was available for that ID, the fallback sequence was the median within Group + student-status, then Group, then the global median. Gender was first copied from other records with the same Occupant ID; if no gender was available for that ID, it was assigned by a seeded random draw from the gender distribution within the same Group, with the overall dataset distribution used as fallback. The original five-category overall IEQ scale (1 = very dissatisfied, 2 = dissatisfied, 3 = ok, 4 = satisfied, 5 = very satisfied) was collapsed into three classes for modeling purposes, with categories 1 and 2 merged as dissatisfied, category 3 retained as neutral, and

categories 4 and 5 merged as satisfied. The distributions on both scales are shown in Figure 6.

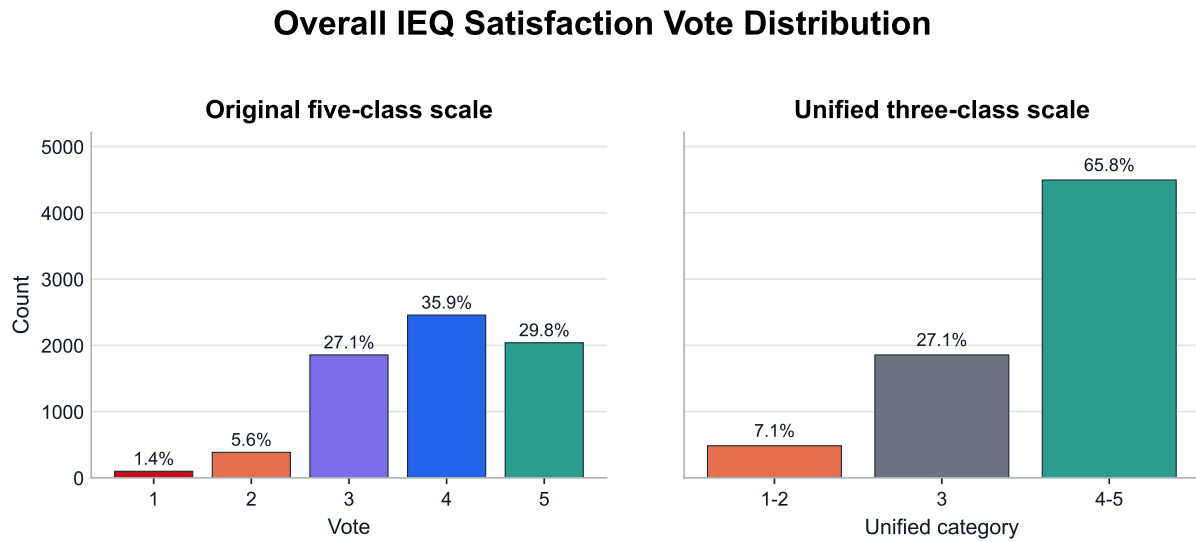


Figure 6: Original and unified overall IEQ satisfaction vote distributions in the current dataset.

Supervisor note: the following supporting figures are intentionally greyed out in the draft to show that they are included only for discussion, orientation, and easier interpretation. They are not yet intended as part of the core article narrative.

3 Analysis

3.1 Modeling Setup

The cleaned dataset was used for ML modelling three-class overall IEQ satisfaction label (dissatisfied, neutral, satisfied). All models were evaluated with five-fold cross-validation using the same folds for fair comparison. The first benchmark stage compared Logistic Regression, Gradient Boosting, ANN/MLP, Random Forest, and Extra Trees. Because the class distribution is imbalanced toward the satisfied class, macro F1 (a score that gives equal importance to all three classes) was used as the main model-selection metric, while raw accuracy, balanced accuracy, and ordinal mean absolute error (MAE) were retained as complementary diagnostics.

3.2 Tuning Workflow

The analysis was done step by step so the draft shows not only the final result, but also how we got there.

1. First, we compared five baseline models on the same feature set.
2. Next, we tuned the strongest models.
3. Extra Trees then received an additional tuning round because it performed best.
4. Finally, we tested several follow-up variants and checked whether the results stayed similar under slightly different settings.

4 Results

4.1 Main Tuned Benchmark

The tuned comparison identifies Extra Trees as the strongest single model for the all-school IEQ benchmark. Extra Trees was the strongest single model in the main benchmark. Random Forest remains the closest competitor, while the simpler linear baseline and the weaker nonlinear baselines lag clearly behind on macro F1.

Table 1: Tuned five-fold cross-validation summary for the current all-school IEQ benchmark.

Model	Accuracy	Bal. acc.	Macro F1	Weighted F1	Ordinal MAE
Extra Trees	0.722	0.592	0.591	0.719	0.316
Random Forest	0.713	0.584	0.582	0.708	0.326
ANN / MLP	0.698	0.495	0.517	0.676	0.345
Gradient Boosting	0.692	0.446	0.464	0.635	0.356
Logistic Regression	0.483	0.464	0.401	0.521	0.671

Extra Trees therefore becomes the current paper-facing reference model. It performs better than the tuned Random Forest on the main benchmark metrics and also improves on the earlier tuned Extra Trees version.

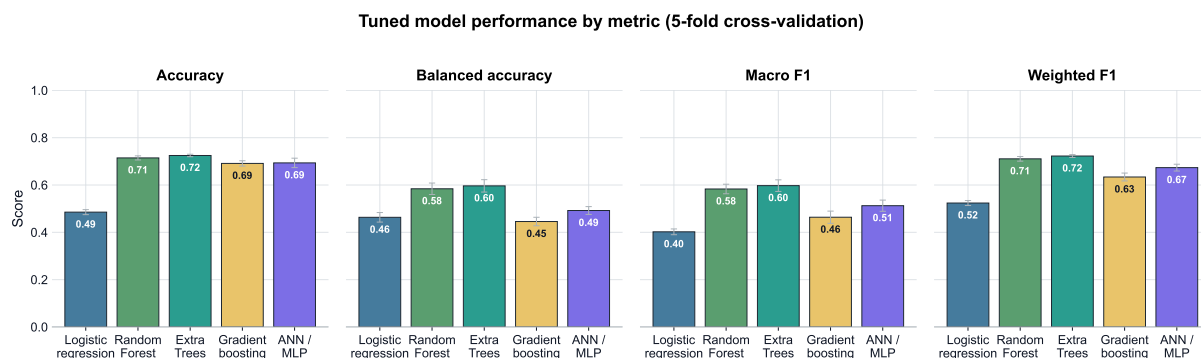


Figure 7: Performance comparison for the tuned IEQ benchmark models under five-fold cross-validation.

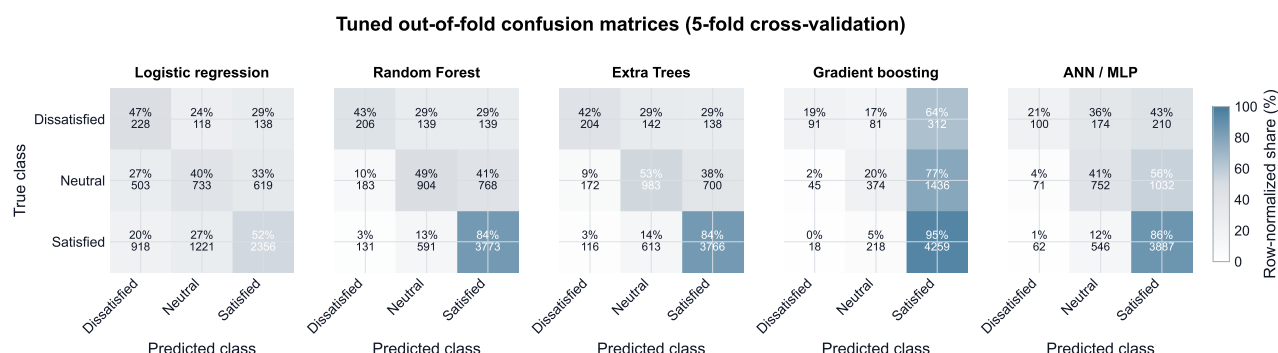


Figure 8: Aggregated confusion matrices for the tuned IEQ benchmark models.

The confusion patterns help explain why macro F1 is a more informative selection metric than raw accuracy in this problem. For the refined Extra Trees model, the satisfied class is recognized strongly, but the model still recovers 42.1% of dissatisfied votes and 53.0% of neutral votes. Random Forest behaves similarly but is slightly weaker for the neutral class. Gradient Boosting and ANN/MLP push too many cases into the satisfied class, which inflates apparent accuracy while damaging minority-class performance. Logistic Regression is the clearest baseline showing that simple linear decision boundaries are insufficient for this classroom IEQ target.

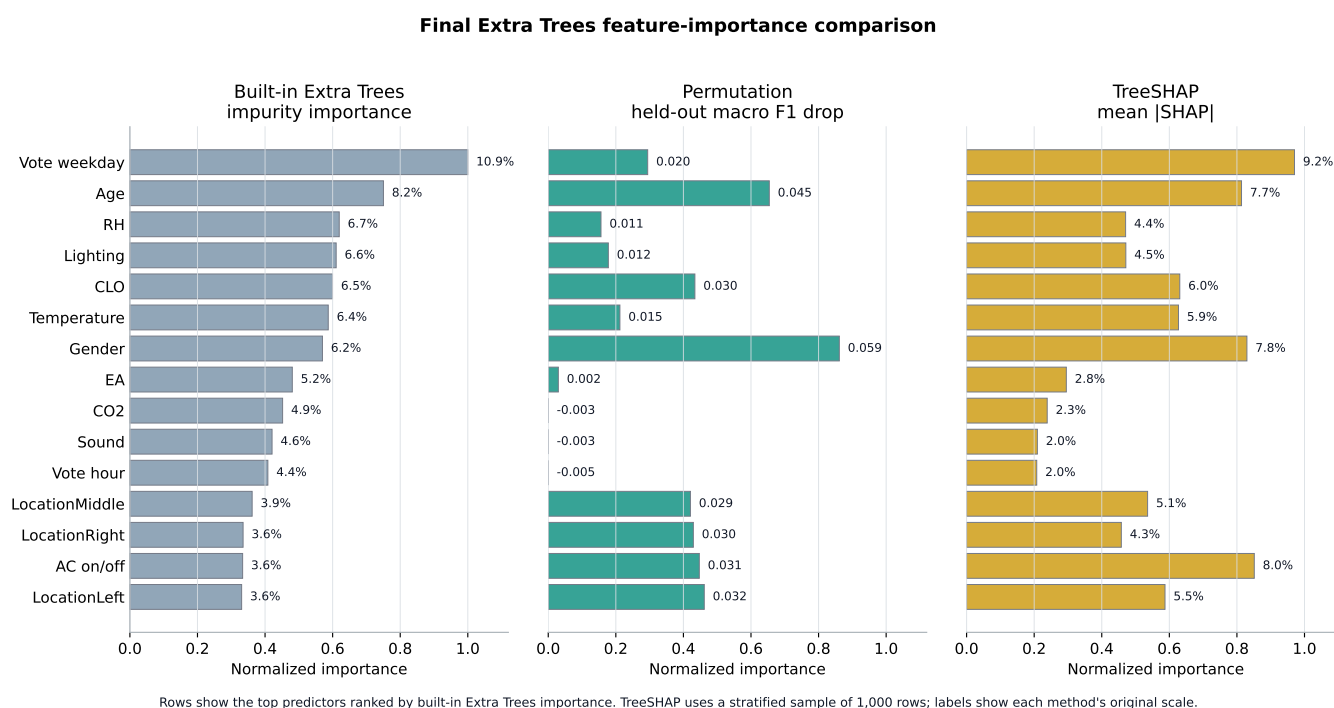


Figure 9: Feature-importance comparison for the final Extra Trees IEQ prediction model.

The feature-importance results vary drastically depending on the method, model or even hyperparameters used. Therefore, the figure is useful for identifying potentially important predictors, but it should not be read as a definitive causal explanation of overall IEQ satisfaction.

4.2 Best Hyperparameters

Table 2: Best tuned configurations for the main IEQ model families.

Model	Selected hyperparameters		
Extra Trees	n_estimators=700, max_samples=0.9, min_samples_leaf=2, class_weight=balanced	criterion=entropy, max_features=None, min_samples_split=10,	bootstrap=True, max_depth=20,
Random Forest	n_estimators=300, min_samples_leaf=4, class_weight=balanced	max_depth=20,	max_features=sqrt, min_samples_split=10,
ANN / MLP	hidden_layer_sizes=(32,), learning_rate_init=0.003	activation=relu,	alpha=0.01,
Gradient Boosting	n_estimators=120, max_features=sqrt,	learning_rate=0.08, min_samples_leaf=3,	max_depth=3, subsample=1.0
Logistic Regression	C=30.0, class_weight=balanced		

4.3 Fine-Tuning Results

Several follow-up experiments were compared with the final selected all-school Extra Trees model.

Manual temperature and CO₂ feature engineering. We added three threshold indicators: CO₂ > 1000 ppm, $T < 18^{\circ}\text{C}$, and $T > 24^{\circ}\text{C}$. Results: accuracy = +0.000, macro F1 = +0.003, ordinal MAE = -0.001. This gave only a very small improvement, so the indicators were not kept as the main model.

Ensemble and ordinal alternatives. The soft-voting ensemble combined model probabilities using 90% Extra Trees and 10% Random Forest. It gave a small gain. Results: accuracy = +0.003, macro F1 = +0.008, ordinal MAE = -0.004. Because the improvement was modest, the single Extra Trees model was kept as the simpler baseline.

Data-quality and row-filtering strategies. We tested complete-case filters that removed rows with missing sensor values before model evaluation. The main-sensor filter kept only rows with complete original Temperature, RH, CO₂, Lighting, and Sound values, leaving 5,349 complete rows. Results: accuracy = +0.004, macro F1 = +0.021, ordinal MAE = -0.006. The stricter filter additionally focused on rows with complete Lighting, Sound, and Ttrend values, leaving 2,314 complete rows. Results: accuracy = -0.012, macro F1 = +0.024, ordinal MAE = +0.008, but it removed many more rows. These filters were not used as the main model because they change the evaluation population and make comparison with the full dataset less direct.

Synthetic class balancing. Class balancing was applied only inside the training folds, leaving validation folds unchanged. We screened three variants: random oversampling to 50% of the majority class, synthetic interpolation to 50%, and synthetic interpolation to full balance. The best balancing variant gave Results: accuracy = +0.006, macro F1 = +0.003, ordinal MAE = -0.000. Because the improvement was small, this approach was not used further.

Primary-school-specific and transfer analysis. Training and testing only on primary-school data improved the primary subset. Results: accuracy = +0.037, macro F1 = +0.033, ordinal MAE = -0.041. But when the primary-trained model was evaluated on all rows, performance dropped. Results: accuracy = -0.020, macro F1 = -0.044, ordinal MAE = +0.032. This suggests domain shift between school types, so primary-only training was not used as the general model.

Table 3: Selected follow-up experiments after the refined Extra Trees model was chosen.

Experiment	Rows	Accuracy	Bal. acc.	Macro F1	Ordinal MAE
Main all-school Extra Trees baseline	6,834	0.722	0.592	0.591	0.316
Threshold T/CO ₂ feature engineering	6,834	0.722	0.596	0.594	0.315
Soft-vote ET 0.90 + RF 0.10	6,834	0.725	0.598	0.599	0.312
Main-sensor complete-case filter	5,349	0.726	0.614	0.612	0.310
Lighting + Sound + Ttrend complete-case filter	2,314	0.710	0.618	0.615	0.324
Best synthetic sampling screen	6,834	0.728	0.597	0.594	0.316
Primary-only ET, full-data model evaluated on primary rows	5,101	0.758	0.611	0.614	0.282
Primary-only ET, trained and evaluated only on primary rows	5,101	0.762	0.625	0.624	0.275

Supervisor note: for a meeting or internal review, I would show the main all-school Extra Trees result first, then immediately show the main-sensor filtered result and the primary-only result as the two most informative sensitivity checks. The ensemble result is interesting, but it is probably too small an increment to be the headline unless the goal becomes purely predictive performance.

Supervisor-only five-class target comparison. I also tried creating a model for 5-class satisfaction with results presented below. I would stick with 3-class classification as results are better and transfer learning will be easier.

Table 4: Supervisor-only comparison of three-class and five-class Extra Trees targets on the 5,240-row complete-case subset.

Variant	Accuracy	Macro F1	Ordinal MAE
3-class	0.725	0.597	0.312
5-class	0.605	0.510	0.540

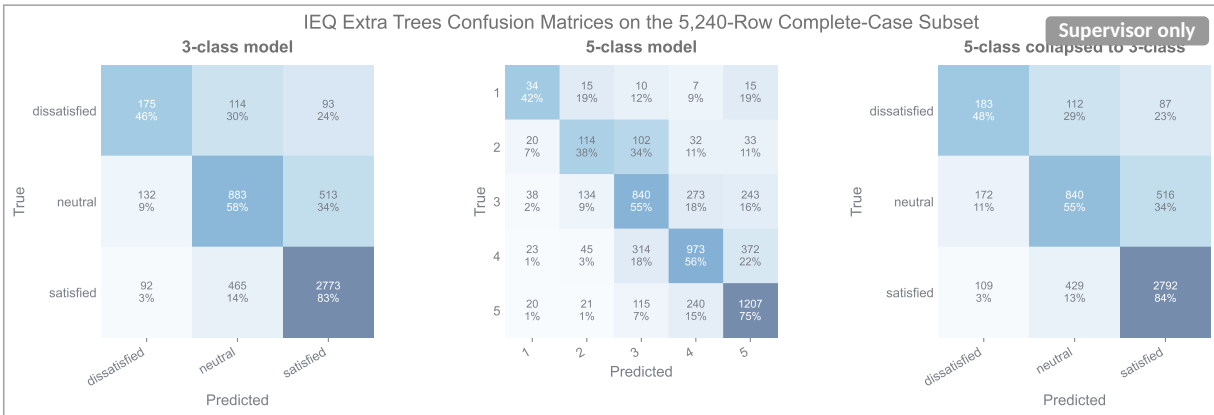


Figure 10: Supervisor-only supporting figure: three-class and five-class Extra Trees confusion matrices on the 5,240-row complete-case comparison subset.

5 Preliminary Literature Benchmark

This section compares my preliminary results with other papers. The selected all-school Extra Trees model reaches accuracy = 0.722, balanced accuracy = 0.592, macro F1 = 0.591, and ordinal MAE = 0.316 for the three-class overall IEQ target. These numbers are not extremely high, but the task is difficult: the model predicts individual classroom votes, across different school settings, with imbalanced classes, and mostly from measured environmental and context data.

Some papers report higher scores, but they often solve an easier or different problem. For example, some use only two classes (satisfied / dissatisfied), train person-specific models, predict an average dissatisfaction rate, or use

subjective votes in other domains (thermal, IAQ, visual, acoustics) as inputs for IEQ prediction. These choices can lead to accuracy above 0.80 or very strong test results, but they are not the same as predicting individual three-class overall IEQ satisfaction from classroom sensors and context variables. Papers that predict overall IEQ from subjective domain satisfaction are still useful, but they start from information that is already close to the final answer [19, 16].

For that reason, the current results seem realistic. It is lower than some values in the literature, but it comes from a harder task. The useful message is that the model works reasonably well on a mixed classroom dataset. The checks also show where the model improves: using only complete sensor rows increases macro F1 by 0.015, and training only on primary-school data improves the primary subset, but does not transfer well to all rows. Therefore I hope that these results are good enough for paper publication.

6 Transfer Learning

I have requested access to the Norwegian classroom dataset, but I have not yet received a response. Based on the available description, this dataset may not include sound information, so the acoustic variables may need to be excluded or treated separately in the transfer-learning analysis.

7 My Questions

Question 1: Should I continue with the three-class target, or should I still try to develop the five-class classification?

Question 2: Are the results good enough to publish?

Question 3: Is there any preferred method to evaluate feature importance?

Question 4: To which journal do we want to submit?

8 Discussion

9 Future Work

Future work should move from retrospective model evaluation toward real-world deployment and operational testing. The current results suggest that overall IEQ satisfaction models may support occupant-centered and energy-aware decision support, but they should not yet be interpreted as ready for direct autonomous building control. A practical next step would be to implement the model in a classroom monitoring dashboard and test whether its risk warnings help school staff identify problematic rooms, lessons, or operating conditions earlier than standard threshold-based monitoring.

Further validation is also needed under real building operation. The model should be tested prospectively in occupied classrooms, where predicted satisfaction can be compared with new student feedback and where actions such as ventilation changes, shading, heating adjustments, or cooling strategies can be evaluated in practice. This would show whether the model is useful not only as a statistical benchmark, but also as a decision-support tool for balancing occupant satisfaction and energy use. Future studies should also examine how well the approach transfers across schools, seasons, and sensor configurations, and whether calibrated risk estimates can be made reliable enough for operational dashboards.

10 Conclusion

11 References

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